AI-Driven Disinformation: Manipulating Social Media Algorithms to Influence Users

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Overview

- Audience: TikTok
- Definition of AI-driven disinformation and its modern relevance
- Why TikTok’s algorithm is uniquely vulnerable

Key Judgements

- TikTok algorithm promotes sensational content
- AI deepfakes manipulate public opinion
- Current moderation efforts are insufficient

Introduction

- AI enables realistic fake media on a large scale
- TikTok's design favors engagement over accuracy
- Disinformation spreads faster than fact-checking can deal with it

Algorithmic Amplification

- Emotion-driven content gets more visibility
- Deepfakes are rewarded by engagement metrics
- TikTok’s design enables rapid misinformation spread

Political Threats

- AI alters perceptions of political figures
- False videos influence election narratives
- Delays in moderation enable impact

Platform Response

- TikTok uses automated and human moderation
- Regional enforcement is inconsistent
- AI detection struggles with nuances

Psychological and Social Effects

- Disinformation fosters mistrust and division
- Users lose faith in institutions
- Repeated exposure skews beliefs

Substantiation

- Brookings and DFRLab documented viral deepfakes in 2024 election cycle
- Pew Research: 45% of users encountered misleading political posts; <20% reported them
- Case example: manipulated videos of 2024 U.S. candidates gaining millions of views

Analysis

Detection lag: creation speed vs. moderation speed mismatch

User media-literacy gap: low reporting rates indicate lack of trust

Regional enforcement disparities and cultural nuances hinder AI detection

Industry Scope

- Social media is a $200 Billion industry
- TikTok has 1.5 Billion global users
- Business models prioritize engagement, not truth

Intelligence Gaps

- Algorithms lack transparency
- Foreign influence campaigns are under-tracked
- Effectiveness of detection tools is unclear

Outlook

- Deepfakes will become more advanced and personalized
- Immersive disinformation (VR/AR) is on the horizon
- Stronger tools and policies are urgently needed

Recommendations

Deploy multimodal AI detectors with cultural and linguistic context

Mandate transparency and accountability

Launch widespread media literacy programs

Conclusion

- AI-driven disinformation threatens democracy and trust
- TikTok’s algorithm amplifies the problem
- Coordinated action is vital to mitigate the threat

Thank you!

References

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